

Activity 0.1 – Everyday Task as an Algorithm

Title: From Daily Routine to Algorithm

Instructions to Students

1. Choose one simple daily activity (examples: making tea, preparing to go to class, booking a cab).
2. Write the steps in clear, numbered points.
3. Make sure each step is:
 - Unambiguous
 - In correct order
 - Does not skip important details

Template

1. Activity chosen: Making Tea
2. Write the steps:
 1. first we have to put water in one bowl
 2. make it to heat & put Tea powder
 3. after that make bowl & put or add the sugar
 4. steep for 2-3 minutes
 5. strain if needed
 6. serve hot (add lemon or normal tea if desired)
3. Check your algorithm:
 - Can a friend follow it without asking questions? Yes / No ☒
 - If "No", refine the steps here:

Activity 0.2 – From Steps to Pseudo-code

Title: Writing Pseudo-code with IF and REPEAT

Instructions

1. Take your steps from Activity 0.1.
2. Identify where decisions happen (IF/ELSE).
3. Identify where repetition happens (REPEAT / WHILE).
4. Rewrite your algorithm using the following keywords:
 - IF, ELSE, ELSE IF
 - REPEAT, WHILE, UNTIL

Template

1. Original activity: Preparation of tea
2. Pseudo-code version:

text

START

IF Tea is very strong THEN

turn off the gas

ELSE

else boil for 3 minutes.

ENDIF

put tea into a cup

REPEAT

Boiling the tea for 3 min

UNTIL minutely.

END

3. Reflection:

- Where did you use IF/ELSE? to check whether the water is boiling or not
- Where did you use REPEAT/WHILE? during steeping time of tea

Activity 1.1 – Run and Edit Given Code

Title: First Python Script – Edit and Run

Instructions

1. Your teacher will give you a small Python script (example below).
2. Do NOT write new code first; only change text or numbers.
3. Run the program and observe the output.

Example Script

python

name = "Student"

age = 18

print("Hello,", name)

print("You are", age, "years old.")

Tasks

1. Change name and age to your own values.

- name: Malush. Patil
- age: 21

2. Run the program. What is the output?

text

Output:

Hello, Malush
You are 21 years old

3. Change the printed message to:

- "Welcome to Python, <your name>"

Write your edited code here:

python

Edited code:

print("welcome to python", name)
print("you are", age, "years old.")
name :- malush. patil
age :- 21

4. Reflection:

- What changed in the output and why?

The output changed because the program now uses the new name & age values, & the printed message was edited.

Activity 1.2 – Predict–Then–Run

Title: Predict Program Output

Instructions

1. Look at each code snippet.
2. BEFORE running, write what you think the output will be.
3. Then run it and compare.

Snippet A

python

x = 5

y = 2

z = x + y * 3

print(z)

- Predicted output: 21
- Actual output: 17

Snippet B

python

a = 10

b = a - 3

a = a + 1

print(a, b)

- Predicted output: (10, 7)
- Actual output: (10, 9)

Reflection

- Where was your prediction wrong? Why?

Activity 2.1 – Calculator in Steps

Title: From Word Problem to Code

Problem 1

A student has marks in 3 subjects. Find total and average.

1. Write the steps in plain language:

1. Write the marks in 3 subject
2. Take the sum of 3 marks, i.e. $\text{total} = a + b + c$
3. To take average divide the total with total number of subjects
4. $\text{Average} = \frac{a + b + c}{3}$

2. Now convert to Python (fill in the blanks):

python

mark1 = 15 $\Rightarrow a$

mark2 = 18 $\Rightarrow b$

mark3 = 20 $\Rightarrow c$

total = 53 $\Rightarrow a + b + c$

average = 17.6 $\Rightarrow \frac{a + b + c}{3}$

print("Total =", total)

print("Average =", average)

3. Test with your own numbers and record output.

$a = 15$ $b = 18$ $c = 20$ $\text{total} = 15 + 18 + 20 = 53$

$\text{Average} = \frac{53}{3} = 17.6$

Activity 2.2 – Fix the Errors

Title: Debugging Simple Python

Instructions

Each snippet has an error.

1. Write what you think is wrong.
2. Correct the code.

Snippet 1

python

print("Hello World)

- Error: Double inverted comma
- Corrected code: Print ("Hello World")

python

undefined

Snippet 2

python

a = 10

b = "5"

c = a + b

print(c)

- Error: b is not a string, double included comma is not used

- Corrected code: a = 10

b = 5

python

print(c)

undefined

Reflection:

- What did you learn about syntax vs logic errors?

Syntax error does not make the program to be executed logical errors are simple & it will affect the output:

Activity 3.1 – Rule-based Classification

Title: Grades with IF/ELSE

Rule

- If mark ≥ 90 : Grade A
- Else if mark ≥ 75 : Grade B
- Else : Grade C

Tasks

1. Draw the decision in simple text:

- Condition 1: if marks is more than or equal to 90 $\rightarrow$ grade A
- Condition 2: if mark is more than or equal to 75 $\rightarrow$ grade B
or else grade C

2. Now write Python code:

python

```
mark = int(input("Enter mark: "))
```

```
if mark > 90:
```

```
    print("Grade A")
```

```
elif mark > 75:
```

```
    print("Grade B")
```

```
else:
```

```
    print("Grade C")
```

3. Test with at least 3 marks and note outputs.

if mark is 95

out put $\rightarrow$ Grade A

mark = 81

output $\rightarrow$ Grade B

mark = 39

out put $\Rightarrow$ Grade C.

Activity 3.2 – Mini Problem Bank

Title: Structured Problem Solving with IF/ELSE

For each question, fill the template.

Template

1. Restate the problem in one sentence:

sowing seeds based on rain/fall prediction (RR)

2. Conditions needed (bullets):

- if rain comes sow the seeds (RR ≥ 500 mm)
- else no sowing of seeds

3. Python code:

python

Your solution

4. Test cases and results:

- Input: _____ Output: _____
- Input: _____ Output: _____

Use this template for:

- Odd or even number.
- Ticket discount if age < 12 or > 60 .
- Check if a year is a leap year (simple rule).

odd or even number

number = int(input("Enter number"))

if number % 2 == 0:

print("Even")

else:

print("Odd")

2) age = int(input("Enter age"))

if age < 12 or > 60 : (i.e. $12 < \text{age} < 60$)

print("Discount in ticket")

else

(print("No discount in ticket"))

Activity 4.1 – Sum with Loops

Title: Sum from 1 to N

Steps

1. In plain language:

- What is "sum from 1 to N"? $1+2+3+4+...+N$

2. Write the algorithm:

- First write / compute $(N+1)$
- Then multiply N with $(N+1)$
- Then divide the product by 2
- Then the quotient value is the sum of number

3. Now code it:

python

```
N = int(input("Enter N: "))
```

```
total = 0
```

```
for i in range(1, N):
```

```
    total = total + i
```

```
print("Sum =", total)
```

4. Try N = 5, 10, 100 and check results.

1) For $N=5$

$$\text{sum} = \frac{5(6)}{2} = \frac{30}{2} = 15$$

2) For $N=10$

$$\text{sum} = \frac{10 \times 11}{2} = \frac{110}{2} = 55$$

3) For $N=100$

$$\text{sum} = \frac{100 \times 101}{2} = \frac{10100}{2} = 5050$$

Activity 4.2 – Star Patterns

Title: Loops for Patterns

Pattern (triangle)

```
*  
**  
***  
****
```

1. Explain the pattern in words:

The pattern is the triangle

2. Write code:

python

```
rows = int(input("Enter number of rows: "))
```

```
for i in range(1, rows + 1):
```

```
    print("*" * i, end=" ")
```

3. Try different rows and observe.

```
row = int(input("Enter number of rows "))
```

```
columns = int(input("Enter number of columns; "))
```

```
for i in range(3)
```

```
for j in range(3)
```

```
    print("*", end=" ")
```

```
    print()
```


Activity 5.1 – Working with Lists

Title: Analyze Marks List

Given

python

marks = [56, 78, 90, 45, 88]

1. In words, write how to find:

- Maximum mark: more than any other mark in a list.
- Average mark: Total marks / 5
- Count of marks ≥ 60 : which is more than or equal to 60

2. Now write Python:

python

marks = [56, 78, 90, 45, 88]

Max

max_mark = _____

Average

total = 0

for m in marks:

total = 56 + 78 + 90 + 45 + 88

average = total / 5

Count ≥ 60

count = 0

for m in marks:

if count $\geq$

count = 3

print("Max:", max_mark)

print("Average:", average)

print("Count ≥ 60 :", count)

Activity 5.2 – Choosing an Approach

Title: Which Method is Better?

Problem

Given a list of 1000 numbers, find how many are positive.

1. Approach A (loop) – describe in words:

for loop

2. Approach B (using built-in functions or list comprehension) – describe:

if else function if no. is > 0 is true or else - u

3. Which is easier to understand for you now? Why?

built in function is easy to understand

4. Implement one approach in Python:

python

numbers = [10, -3, 5, 0, 8, -2] # example

Your code

```
number = int(input("Enter number"))
```

```
if number > 0:
```

```
    print("number is positive")
```

```
else:
```

```
    print("number is negative")
```


Activity 6.1 - Creating a Function

Title: Wrap Logic into a Function

Task

Turn average-of-marks logic into a function.

1. In words: what does the function do?

2. Write code:

python

```
def average_marks(marks_list):
```

```
    total = 0
```

```
    for m in marks_list:
```

```
        total = total + marks
```

```
    avg = total / m
```

```
    return avg
```

```
data = [60, 70, 80]
```

```
print("Average =", average_marks(data))
```

3. Test with another list and note output.

```
data = [10, 20, 15]
```

```
Print ("Average =", average_marks(data))
```

Out put:

Average = 15

Activity 6.2 – Problem-Solving Template

Title: Standard Template for Any Problem

Use this for any new problem.

1. Problem understanding

- What is given (inputs)?

age is given

- What is required (outputs)?

whether discount is applicable or not

- Any constraints (limits, ranges)?

btw 12 - 60 years of age no discount applicable

2. Analysis

- Write steps in bullets (algorithm):

- if age is < 12 50% discount
- elif age is > 60 30% discount
- else no discount

3. Design

- Will you use loops? Which? no
- Will you use conditions (if/else)? yes
- Will you use functions? Which ones? no
- What data structures (list, etc.)? no

4. Code (Python)

python

Your implementation

5. Test

- Test case 1: input _____ → output _____
- Test case 2: input _____ → output _____
- Edge case: input _____ → output _____

Activity 7.1 – Capstone Problem

Title: Simple Fee Calculator

Problem

A class charges a base fee.

- If student is below 12 years, 50% discount.
- If above 60, 30% discount.
- Otherwise, no discount.

Use the full template from Activity 6.2 here.

(Attach a blank copy of the template and ask students to fill all five sections.)

```
age = int(input("Enter age"))
```

```
if age < 12:
```

```
    print("50% discount")
```

```
elif age > 60:
```

```
    print("30% discount")
```

```
else:
```

```
    print("no discount")
```


Activity 7.2 – Reflection Sheet

Title: How I Solve a New Problem

1. When I see a new problem, my first step is:

Problem understanding → check input, outputs + limits / ranges if in case present

2. Before coding, I make sure I know:

- Inputs: _____
- Outputs: _____

3. The tools I think about (loops, if, functions) are:

Loops- for- while, if else data structure list, set dictionary, tuple etc.

4. One thing I will do differently from now on:

clearly understand the program problem to be solved.